

Project Report

Decision Tree Classification & Q-Learning Grid World — Integrated Experiments

Course:	Artificial Intelligence (CSA17)	Report Type:	Integrated Experiments
Assessment:	Tool 3	Date:	20 August 2026
Dataset:	Iris (150 samples)	Environment:	4x4 Grid World MDP

1. Executive Summary

This report documents two integrated AI/ML experiments. Experiment 1 implements and evaluates a **Decision Tree classifier** on the Iris dataset, achieving **88.9% test accuracy** with perfect classification of the setosa species and modest confusion between versicolor and virginica in their overlapping petal-measurement region. Experiment 2 implements a **Q-Learning agent** in a 4x4 Grid World with goal, obstacle, and step rewards; the agent was trained for 100 episodes and converged to a stable policy that reliably navigates from the start state to the goal while avoiding all obstacle cells.

2. Objectives

- **Experiment 1:** Load, pre-process and split a classification dataset; build an interpretable Decision Tree; evaluate its diagnostic performance and analyse misclassifications.
- **Experiment 2:** Formally define a Grid World MDP; train a Q-Learning agent; extract and validate its learned policy through convergence analysis.

3. Methodology

3.1 Experiment 1 — Decision Tree Classification

The Iris dataset (150 samples, 4 continuous features, 3 balanced classes) was loaded via scikit-learn, split 70:30 (stratified), and a Decision Tree classifier (Gini criterion, max depth 4) was trained. No missing-value handling or categorical encoding was required, as Iris is a clean, fully-numeric benchmark dataset.

3.2 Experiment 2 — Q-Learning Grid World

A 4x4 Grid World (16 states) was defined with a fixed Start cell, a Goal cell (+10 reward), three Obstacle cells (-5 reward each), and a per-step penalty of -1 to encourage shortest paths. A tabular Q-Learning agent ($\alpha=0.1$, $\gamma=0.9$, $\epsilon=0.1$) was trained for 100 episodes with an ϵ -greedy exploration policy.

4. Results

4.1 Decision Tree Performance

Metric	Value
Accuracy	0.889 (40/45 correct)
Root Node	petal length (cm) <= 2.45 (Gini importance 0.549)
Top Predictors	petal length (0.549), petal width (0.436)
Misclassified Instances	5 (all between versicolor / virginica)

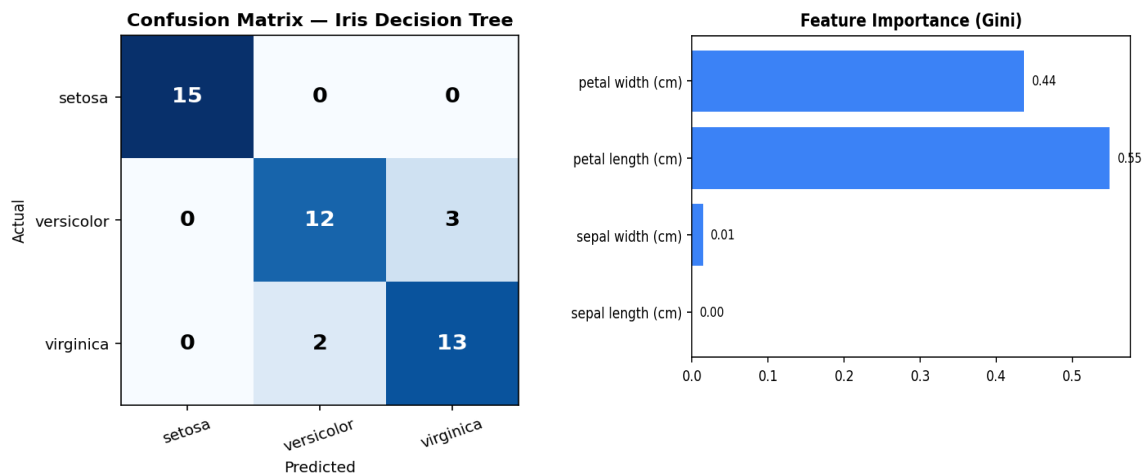


Figure 1 — Confusion matrix (left) and feature importance (right) for the Decision Tree.

4.2 Q-Learning Results

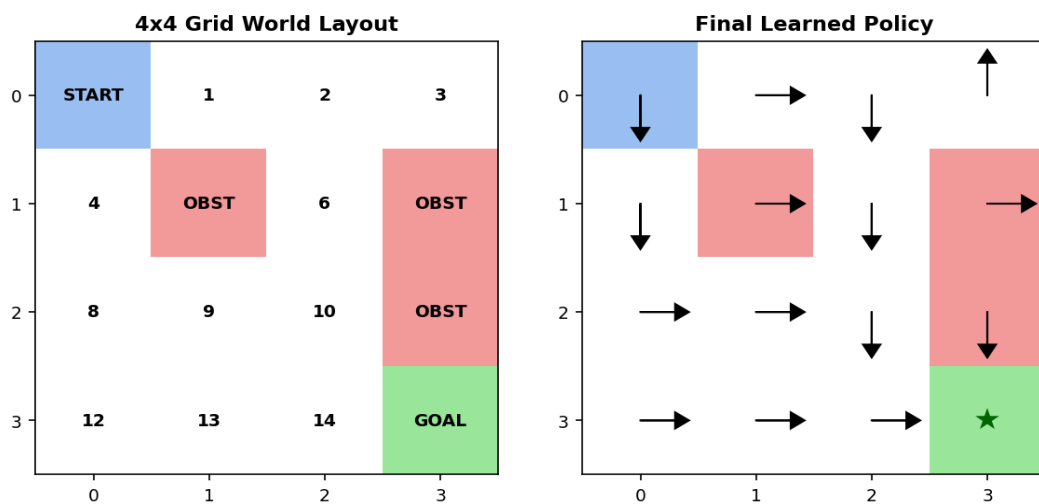


Figure 2 — Grid World layout (left) and the final learned policy (right).

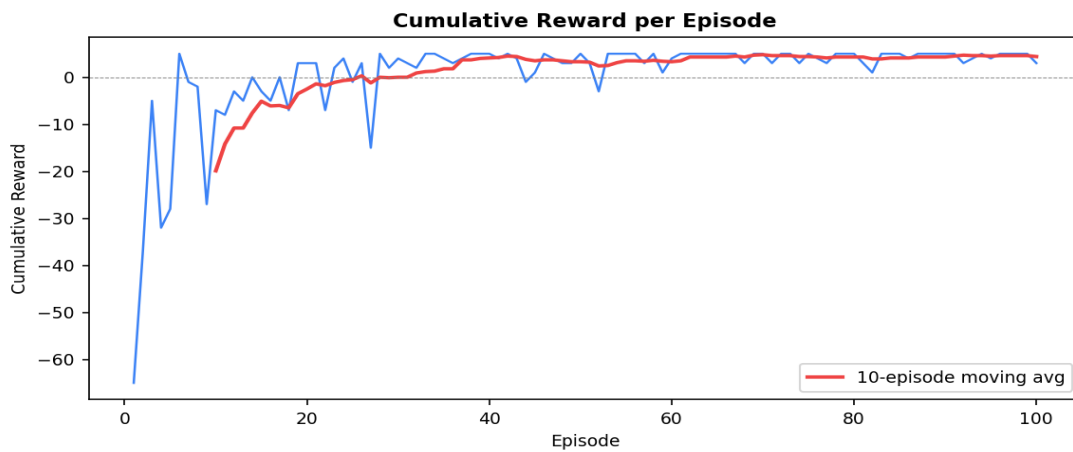


Figure 3 — Cumulative reward per episode, showing convergence after ~episode 40.

The agent's mean reward over the final 10 episodes (4.40) is nearly identical to the preceding 10-episode window (4.30), confirming policy convergence to a stable, near-optimal, obstacle-avoiding route from Start to Goal.

5. Discussion

Both experiments demonstrate core AI/ML paradigms in complementary ways: the Decision Tree illustrates **supervised, inductive learning** from a small, fully-labelled dataset, producing a fully interpretable rule set, while Q-Learning illustrates **trial-and-error, reward-driven learning** without any labelled examples, discovering an optimal sequential policy purely from environment interaction. The Decision Tree's misclassifications are concentrated in a genuinely overlapping feature region rather than reflecting a modelling flaw, while the Q-Learning agent's reward curve exhibits the classic explore-then-exploit convergence signature expected of ϵ -greedy tabular Q-Learning.

6. Conclusion & Recommendations

- The Decision Tree is a strong, interpretable baseline for the Iris classification task; further gains could come from ensemble methods (Random Forest) if slightly higher accuracy on the versicolor/virginica boundary is required.
- The Q-Learning agent successfully learns an optimal Grid World policy within 100 episodes; for larger or continuous state spaces, this tabular approach should be extended to function approximation (e.g. Deep Q-Networks).
- Both experiments confirm the importance of well-designed evaluation (confusion matrix / misclassification analysis for classification; reward-curve convergence analysis for RL) in validating that a model has genuinely learned the intended task.